

1. SAMPLE & TEST INFORMATION

Field	Result
Laboratory ID	TB-TNGS-DEMO-001
Specimen type	Sputum
Test	TB targeted next-generation sequencing (tNGS)
Sequencing platform	Illumina NextSeq
MTB detected	DETECTED
Analysis status	PASS
Report type	Demonstration

2. DRUG-RESISTANCE SUMMARY

Drug	Relevant variant	Interpretation
Rifampicin	<i>rpoB</i> S450L	Resistant
Isoniazid	<i>katG</i> S315T	Resistant
Levofloxacin	<i>gyrA</i> D94G	Resistant
Moxifloxacin	<i>gyrA</i> D94G	Resistant
Amikacin	No validated resistance mutation	No resistance detected
Bedaquiline	No validated resistance mutation	No resistance detected

3. RESISTANCE-ASSOCIATED MUTATIONS

Gene	Mutation	Associated drug(s)	Allele frequency	Read depth	Interpretation
<i>rpoB</i>	S450L	Rifampicin	98.7%	1,842x	Resistance associated
<i>katG</i>	S315T	Isoniazid	99.2%	2,105x	Resistance associated
<i>gyrA</i>	D94G	Levofloxacin; Moxifloxacin	97.6%	1,764x	Resistance associated

Interpretation source: WHO Catalogue of mutations in *Mycobacterium tuberculosis* complex and their association with drug resistance.

4. SEQUENCING & ANALYSIS QUALITY CONTROL

QC indicator	Number	Percentage	Status
Total sequencing reads	1,842,516	100.0%	PASS
Reads passing QC	1,731,964	94.0%	PASS
Reads mapped to H37Rv	1,693,022	97.8% of QC-passing reads	PASS
Resistance targets adequately covered	12 / 12	100.0%	PASS
Resistance-associated mutations detected	3	N/A	-
Overall sequencing and analysis QC	-	-	PASS

Reference genome: *Mycobacterium tuberculosis* H37Rv (NC_000962.3)

QC assessment: Sequencing, mapping and target coverage met the demonstration requirements for downstream resistance-mutation interpretation.

5. FINAL INTERPRETATION

DRUG-RESISTANT TB PROFILE PREDICTED

Targeted NGS detected resistance-associated mutations in *rpoB*, *katG* and *gyrA*, predicting resistance to **rifampicin, isoniazid, levofloxacin and moxifloxacin**.

No validated resistance-associated mutation was detected for amikacin or bedaquiline in the reported, adequately covered targeted regions.

Results should be interpreted according to the validated tNGS assay, the WHO mutation catalogue, applicable national TB diagnostic algorithms, clinical information and, where required, complementary phenotypic drug-susceptibility testing.

Report Metadata

Reference genome: H37Rv NC_000962.3

Interpretation resource: WHO TB Mutation Catalogue

Pipeline status: PASS